Two-Point Correlations: Centered Graphs, Sharper Spectral Transfer, and Finite Weighted Error Certificates

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Abstract

We study selected applicable steps of the September 24, 2026 manuscript *Ordinary two-point correlations of multiplicative functions*. The whole eighteen-file source structure is mapped, while literal reading and verification are separately delimited. A parent-supplied resolvent refinement is proved for noncommuting self-adjoint edge operators: with single-edge bound a, compressed square bound b^2, and nonbacktracking spectral radius r, the compressed adjacency norm is at most $c + b^2/(c-a)$, where $c = \max(r, a+b)$ and $b > 0$. Further local results give energy-weighted moment transfer without independence, displacement-mass boundary bounds, exact singleton-centering sensitivity, and finite polynomial-compression and arithmetic-energy certificates. The private graph-transform and finite Gram formulations are used only in their finite-dimensional domains. The rank-witness, coefficient-uniform forest, short-interval, and every-cutoff arithmetic obligations remain source inputs; neither a new Chowla theorem nor a full proof audit is claimed. Contribution assessment: C2/5, because the proved local refinements sharpen and clarify finite transfer estimates without new arithmetic cancellation or a stronger global theorem.

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1 Scope, Attribution, and the Actual Target

This is report 013 in the separately authorized first-51 L3 stage, not a revision of any earlier report. Applicability L3 measures suitability of a method; C2 measures the realized contribution of this report. Length, transformed notation, and imported source theorems do not raise the latter score.

The source [1] states that one absolute $c > 0$ works for every fixed $a_1, a_2 \geq 1$, $b_1, b_2 \geq 0$, with $a_1 b_2 - a_2 b_1 \neq 0$:

$$\left| \sum_{1 \leq n \leq X} \lambda(a_1 n + b_1)\lambda(a_2 n + b_2) \right| \ll_{a_1,a_2,b_1,b_2} X(\log X)^{-c} \quad (X \geq 3). \tag{1}$$

Here X is any real cutoff, $\lambda(n) = (-1)^{\Omega(n)}$, and Ω counts prime factors with multiplicity. Its qualitative assertion concerns multiplicative $f_i : \mathbb{N} \to \{z \in \mathbb{C} : |z| \leq 1\}$, not necessarily completely multiplicative, with at least one *original* factor satisfying, for every fixed Dirichlet character χ,

$$\inf_{|t| \leq N} \sum_{p \leq N} \frac{1 - \Re(f_i(p)\overline{\chi(p)p^{it}})}{p} \longrightarrow \infty. \tag{2}$$

The prime cutoff and frequency bound are both N. The assertion is ordinary Cesaro cancellation, including fixed progressions and fixed nonproportional affine forms. This report does not upgrade a logarithmic average or a result outside exceptional scales into that assertion.

Status vocabulary. *Source* means a statement or argument in the assigned manuscript; *local* means a finite result proved below; *conditional assembly* means a consequence only if the named source inputs are supplied with their stated uniformities. The optimized transfer in Theorem 4.1 was supplied by the parent as a new local deduction during this batch and is credited accordingly. The proof here checks it against the source lemma. It is neither a theorem imported from the private corpus nor a historical novelty claim. Other local certificates below are elementary extensions or reformulations, not a replacement for the source's arithmetic work.

Read scope and identity. The assigned 18 TeX files contain 6,192 lines; 4,089 distinct lines were literally displayed and read in bounded, untruncated chunks. All section, subsection, and input structure was mapped. The two actual private originals contribute 695 distinct lines out of 908. Reading a proof is not its independent formal verification. No whole-corpus reading is asserted. The bibliography was also read as a supplemental bibliographic file, separately from these 18-file counts.

The parent's authenticated identity record covers 20 public texts, adding that bibliography and the source README; it reports byte identity from the pinned base to the captured October 8 target. It does not certify arithmetic validity or a later live branch.

2 Whole-Source Map and Read Boundaries

Paths in this table are relative to the source's `build/` directory; `Q/` abbreviates `quantitative/` and `L/` abbreviates `qualitative/`. "Full" means literal file reading only. Exact individual read calls, endpoint SHA256 hashes, and the full heading map are in the accompanying read-scope ledger.

File	Lines read	Mathematical role and remaining boundary
`main.tex`	1–90, full	Assembly, abstract, three-part order.
`introduction.tex`	1–278, full	Exact targets, prior work, two distinct limit procedures.
`Q/01-setup.tex`	1–199, full	Prime supplies, padding, bins, independent residues.
`Q/02-finite-law.tex`	1–239, full	Bit encoding, exact-independence correction, truncated weights; external circuit theorem imported.
`Q/03-analytic.tex`	1–40, 277–329, 381–567	Centering statement, padding definition, deletion proof; rough Fourier and most padding analysis not read.
`Q/04-paths-trace.tex`	1–421, full	Prohibited paths, edge matrices, signed trace expansion, mixed differences.
`Q/05-rank-witness.tex`	1–492, full	Real-rank estimate, common-hybrid witnesses, ordered elimination. Read, not independently certified globally.
`Q/06-forest.tex`	1–378, full	Coefficient-independent forest code, common-environment padding sum, trace completion. Same certification limit.
`Q/07-spectral.tex`	1–333, full	Noncommuting transfer, row squares, testing, prefix boundaries, absolute constants.
`Q/08-affine.tex`	1–51, full	Exact progression-to-affine endpoint identity.
`L/graph-setup.tex`	1–120, full	Core/center bands, $\theta = 1/4$, W, graph estimate.
`L/analytic-transfer.tex`	226–436	Centering statement, contradiction, finite-prime stability and short-sum input; bias construction and rough multiplier proof unread.
`L/paths.tex`	1–125	Specifications, deleted set, high-trace statement; subsequent density/arithmetic proof unread.
`L/thinning.tex`	1–104	First visits, retained configurations; connected-block and corruption estimates unread.
`L/cylinder-sieve.tex`	144–238	Conditional reduction and witness statement; poset proof and remaining elimination unread.
`L/trace-count.tex`	1–115, 220–437	Tree weights, signed prime integration; record enumeration and final prime summation unread.
`L/localization.tex`	1–167, full	Directed singular moments, discarded edges, bilinear testing.
`L/affine-corollaries.tex`	1–173	Finite dilation/residue expansions; progression assembly and specializations unread.

The catalogue path for TR190-GGRT resolves to the actual file in the `TRs/TR190-GGRT` directory. A separately found duplicate has the same hash and is not counted twice. Its lines 1–277 and

303–342 were read. TR1674's actual `TR1674-revised-v2.tex` was read at lines 65–378 and 430–493. The numerical appendices and unlisted private passages were not audited. Full paths and hashes, including the resolved originals, are local evidence, not public search inputs.

3 The Quantitative Centered Operator

We preserve the source normalization instead of treating the graph as an unweighted adjacency matrix. Fix integers $h, l \geq 1$ and $b_0 \in \mathbb{Z}$. Choose absolute constants A first and $W \geq 10$ later, and set

$$L = (\log X)^{1/A}, \quad \delta = 1/200, \quad J = \left\lfloor \frac{\delta \log L}{6W} \right\rfloor, \quad \eta = e^{-J}, \quad K = e^{4J}, \quad k = \lfloor L \rfloor.$$

We work above the threshold where $J \geq 1$. The disjoint supplies P_i consist of primes $p \equiv 1 \pmod 5$, $p \nmid lh$, with $L^{1-\delta} e^{6W(i-1)} \leq \log p < L^{1-\delta} e^{6Wi}$. The padding set Q contains the other primes up to e^L, excluding divisors of lh. Write

$$V_i = \sum_{p \in P_i} p^{-1} \in [W, 2W], \quad V_* = \prod_i V_i, \quad \mathcal{D} = \{p_1 \cdots p_J : p_i \in P_i\}.$$

For squarefree Q-products, including 1,

$$u(q) = 4^{\omega(q)}, \quad w(n) = \sum_{q \mid n} u(q) = 5^{\omega_Q(n)}, \quad S = \prod_{p \in Q} (1 + 4/p). \tag{3}$$

The bin $\mathcal{S}_j$ imposes $\omega(q) \leq 100 \log L$ and $j\eta \leq \log(dq) < (j+1)\eta$, with $j\eta \leq 100L$; $T_j = Xe^{j\eta}$ and $S_0 = \sum_{j,(d,q) \in \mathcal{S}_j} u(q)/(dq) \geq SV_*/2$. At each translated site use the *same* residue coordinate for each prime. The auxiliary product law is not independent resampling at different sites.

Let $\rho_j(n,d) = w(n)^{-1} \sum_{q \mid n,\, j\eta \leq \log(qd) < (j+1)\eta} u(q)$. It includes all squarefree padding divisors in that bin, before the $\omega(q)$ restriction. The source predicate σ_j deletes starts of positive minimal prohibited words. Define

$$\tau_j(n,d) = \mathbf{1}_{\omega_Q(n) \leq 400 \log L} \mathbf{1}_{\rho_j(n,d) \leq K/L} \sigma_j(n), \qquad E = \operatorname{diag}(\mathbf{1}_{\omega_P(n) \leq 6WJ}).$$

The P-degree cutoff is deliberately omitted from τ_j. On $I_t = \{t+1, \ldots, t+M\}$, $M = \lceil e^{103L} \rceil$, define real symmetric matrices, for $n' = n + hqd$,

$$A_d(n,n') = \frac{Lu(q)\tau_j(n,d)\tau_j(n',d)}{\sqrt{w(n)w(n')}} \mathbf{1}_{q \mid n} \mathbf{1}_{n \equiv b_0 qd \pmod l} \prod_{p \mid d} (\mathbf{1}_{p \mid n} - 1/p). \tag{4}$$

Set the reverse entry equal, and other entries to zero. On the d-indexed direct sum of block spaces set $H_{d,d'} = \mathbf{1}_{d \neq d'} A_{d'}$. This H is generally nonnormal. A spectral theorem for self-adjoint adjacency cannot be applied to H as if it were normal.

3.1 The arithmetic input that cannot be transformed away

The source's moment, with *unnormalized* Hilbert–Schmidt norm, is

$$\mathbb{E} \|H^k\|_{\mathrm{HS}}^2 \leq R^{2k}, \qquad R = K(C_2 \sqrt{W})^J. \tag{5}$$

The integer-origin version is uniform over every interval of length at least $\exp(L^A/2)$. An absolute C_2 must be fixed before choosing W. In particular, there is no omitted dimension divisor in this formula.

The read source chain has the following indispensable structure.

1. Closed words of length $2k$ arise by joining two paths; nonbacktracking holds within each half, not necessarily at the two joins. Repeated vertices are counted with multiplicity, so all square-root denominators become $\prod_i w(n+\xi_i)$. The external copy and site counts remain in the moment.

2. Singleton prime integration is performed before absolute values. For uniform $Z \pmod p$, $\mathbb{E}[(\mathbf{1}_{Z=a} - 1/p)F(Z)] = p^{-1}\mathbb{E}[F(a) - F(Z)]$. Deleting vertices introduces a mixed difference of the common-hybrid survival predicate. For nonsingletons, lit tests force one residue; unlit factors are scalars $-1/p$, not nondivisibility tests.

3. High real rank gives independent control/minor variables after fixing other columns and padding. Real nonsingularity is not modular invertibility. The source random-prime bound subtracts two independent copies to remove the control-dependent term and counts divisors of nonzero integer coordinates.

4. Many singletons force simultaneously positive witnesses in one hybrid. The private label of a witness need not have an invertible coefficient; the active-label lemma and ordered prefix comparisons provide that coefficient. Selected variables must be eliminated in reverse dependency order.

5. For the remaining words, regular directions in a quotient are independent. Their line/point incidence graph is a forest, and each regular segment is decoded from its endpoint lines. The code counts the *union* over all numerical coefficients and padding choices, not just one fixed realization.

6. Only then can padding be summed backwards in one common residue environment, using a departure mass bound K at every site, giving K^{2k}. The distinct-label count $m_P \le Jk + S_1/2$ produces the square-root mass gain.

These arguments were read in quantitative files 04–06; no independent global audit of their enumeration or all arithmetic subcases is asserted. The finite-law proof was read, but Braverman's theorem and the analytic centering input are imported. In particular, an unrestricted integer average of w^2 is not supplied by the truncated finite-law comparison.

4 Optimized Noncommuting Transfer

All inner products are conjugate-linear in the first argument. Operator norms are Euclidean norms on finite-dimensional complex Hilbert spaces.

Theorem 4.1 (Parent-supplied local refinement)**.** *Let $B_1, \dots, B_N$ be self-adjoint on $\mathcal{H}$, $N \ge 1$, and let P be an orthogonal projection. Suppose*

$$\|B_i\| \le a, \qquad \sum_i \|B_i v\|^2 \le b^2 \|v\|^2 \quad (v = Pv), \qquad a, b \ge 0.$$

On $\mathcal{H}^N$ define $\mathscr{B}_{ij} = \mathbf{1}_{i\ne j} B_j$ and $r = \rho(\mathscr{B})$. If $b = 0$, then $P(\sum_i B_i)P = 0$. If $b > 0$, then

$$\|P(\sum_i B_i)P\| \le \inf_{0<t<1/\max(a,r)} \left(\frac{1}{t} + \frac{b^2 t}{1-at}\right) =: \Psi(a,b,r) \tag{6}$$

$$= \begin{cases} a + 2b, & r \le a + b, \\ r + \dfrac{b^2}{r-a}, & r \ge a + b. \end{cases} \tag{7}$$

When $\max(a,r)=0$*, the interval in* (6) *means* $(0,\infty)$*. Equivalently* $\Psi = c + b^2/(c-a)$ *with* $c=\max(r,a+b)$*. No commutation between distinct* B_i*, or between* P *and* B_i*, is assumed.*

Proof. For $|u| < 1/\max(a,r)$ both $I - u\mathscr{B}$ and $I + uB_i$ are invertible. Define

$$F(u) = I - \sum_i uB_i(I+uB_i)^{-1}.$$

Each summand is self-adjoint. If $F(u)v = 0$, set $z_i = (I+uB_i)^{-1}v$. Then $(u\mathscr{B}z)_i = v - uB_iz_i = z_i$; a nonzero v would contradict invertibility of $I - u\mathscr{B}$. Thus $F(u)$ is invertible, and continuity from $F(0) = I$ implies positivity throughout that real interval. This uses spectral radius to exclude eigenvalue $1/u$, not a norm-convergent series for the generally nonnormal $\mathscr{B}$.

The exact identity

$$F(u) = I - u\sum_i B_i + u^2 \sum_i B_i^2 (I+uB_i)^{-1}$$

and the spectral calculus of each individual B_i give, for $v = Pv$,

$$0 \le \sum_i \langle B_i v, (I+uB_i)^{-1}B_i v\rangle \le \frac{b^2}{1-a|u|}\|v\|^2.$$

Testing positivity at $u = t$ and $u = -t$ proves (6). For optimization put $x = 1/t > \max(a,r)$. The objective is $x + b^2/(x-a)$, whose derivative is $1 - b^2/(x-a)^2$ and whose minimum on $x > a$ occurs at $x = a+b$. If $r \ge a+b$, its infimum on the allowed open interval is the limit as $x \downarrow r$. At $r = a+b$ the same limiting argument applies. No endpoint inverse is asserted to exist. This proves (7). If $b = 0$, every $B_iv = 0$ for $v = Pv$, directly giving the claimed zero compression. □

Corollary 4.2 (Comparison and source substitution)**.** *For* $b > 0$*,* $\Psi \le 3\max(a,b,r)$*. It is nondecreasing in* r *and in* b*. On each source block one may take*

$$a = 2K, \qquad b = 2K(8W)^{J/2}. \tag{8}$$

If (5) *holds on an origin average, then outside a fraction* e^{-2k} *of origins one may replace the source's* $3\max(a,b,eR)$ *by* $\Psi(a,b,eR)$*.*

Proof. For $r \le a+b$, $a+2b \le 3\max(a,b,r)$. Otherwise $b \le r-a$, so $\Psi \le r+b \le 2r$. Monotonicity follows either from the infimum or the two matching branches; the derivative in r on the second branch is $1 - b^2/(r-a)^2 \ge 0$. Finally $\rho(H)^{2k} \le \|H^k\|_{\mathrm{HS}}^2$ and Markov's inequality apply even when H is nonnormal. The source's row-square bounds give (8). □

For illustration, $(a,b,r) = (1,10,0)$ changes the scalar bound from 30 to 21; $(1,10,100)$ changes it from 300 to $100 + 100/99$. These are parameter examples, not computed arithmetic blocks. This refinement can matter for finite constants but does not establish a larger admissible global Chowla exponent. The source has not supplied a numerical value of C_2 here.

4.1 A block-specific square certificate

Put $a_d(n) = \prod_{p|d} |\mathbf{1}_{p|n} - 1/p|$. It is constant along each edge of A_d. The source weighted row argument proves

$$\|A_d v\|^2 \le 4K^2 \sum_n a_d(n)^2 |v_n|^2.$$

Consequently a directly certified replacement is

$$b_t^2 = 4K^2 \max_{n:\, E_{nn}=1} \sum_d a_d(n)^2 \le 4K^2(8W)^J, \tag{9}$$

where the maximum of an empty retained set is zero. The proof is summation of the displayed diagonal majorants. Using b_t changes neither the trace matrix nor the deletion event. Replacing the entire source proof by numerical evaluation of these enormous blocks is not a proposed efficient algorithm.

5 Weighted Moment Transfer Without Independence

The exceptional-block event and the weight of the test vector generally correlate. The following local extension keeps that dependence unrestricted.

Theorem 5.1 (Energy-weighted tail certificate)**.** *On a finite probability space of blocks suppose Theorem 4.1 holds with common $a \ge 0, b > 0$. Let $r_t = \rho(\mathscr{B}_t)$, $g_t = P_t g_t$, $\alpha_t = \|g_t\|^2$, and assume, for $p > 1$,*

$$\mathbb{E} r_t^p \le R^p, \qquad \mathbb{E}\alpha_t \le A_0, \qquad 0 \le \alpha_t \le T_0,$$

where $A_0, T_0 > 0$ are certified upper bounds. Put $d = a + b$, $s_0 = R(T_0/A_0)^{1/p}$, and $z = \max(d, s_0)$. Then

$$\mathbb{E}\left|\langle g_t, \sum_i B_{i,t} g_t\rangle\right| \le (a+2b)A_0 + A_0(s_0 - d)_+ + \frac{T_0 R^p}{p-1} z^{1-p}. \tag{10}$$

In particular the last two terms together are at most $\frac{p}{p-1} R A_0^{1-1/p} T_0^{1/p}$. For $R = 0$ they are interpreted as zero. If $b = 0$ on all blocks, the form vanishes and no tail estimate is needed.

Proof. For $r \le d$ use $\Psi = a + 2b$. For $r > d$, $b^2/(r-a) \le b$, giving $\Psi \le a + 2b + (r-d)_+$. Also, without independence,

$$\mathbb{E}[\alpha_t \mathbf{1}_{r_t > s}] \le \min(A_0, T_0 \mathbb{P}(r_t > s)) \le \min(A_0, T_0 R^p s^{-p}).$$

The layer-cake identity bounds $\mathbb{E}[\alpha_t (r_t - d)_+]$ by the integral of this minimum over $s \in [d, \infty)$. Split at s_0 to obtain (10); integration from zero instead gives the simpler bound. If $R = 0$, $r_t = 0$ almost surely. All steps use positive integrands. □

Source-specific consequence, conditional on the source moment. For the quantitative source take $p = 2k$, $g_t = E(\sqrt{w}\, \lambda \mathbf{1}_{\omega_Q \le 400 \log L})$, and

$$A_0 = 2MS, \qquad T_0 = M 5^{400 \log L}.$$

The first bound follows from the *truncated* weight comparison; the second is deterministic. Divide (10) by $2LM$ and add the prefix error in Theorem 6.1. The simpler version gives the retained-bin estimate

$$|C_j^\circ| \le \frac{S}{L}\left[a + 2b + \frac{2k}{2k-1} R \left(\frac{5^{400 \log L}}{2S}\right)^{1/(2k)}\right] + \mathcal{E}_{\text{prefix}}. \tag{11}$$

No exceptional blocks have been silently discarded. The formula remains valid if the ratio inside parentheses is less than one. For fixed W, its logarithm divided by $2k$ is $O(\log L/L)$ in absolute value: $1 \le S \le 5^{|Q|}$ alone is not the useful estimate, but the source prime harmonic bounds give

$\log S = O(\log L)$. Thus the multiplier of R is $1 + o(1)$. This trades the source's fixed Markov threshold for an integrated tail bound and an additive $a + 2b$ term; it need not dominate the optimized threshold bound for every finite parameter set. Both bounds may be retained and their minimum used.

6 Boundary, Deletion, and Truncation Certificates

The window multiplicity is an exact combinatorial count, independent of multiplicativity. This gives a refinement sensitive to the actual distribution of edge lengths.

Theorem 6.1 (Displacement-mass prefix comparison)**.** *Let $T \geq 1$, $N = \lfloor T \rfloor$, and integers $1 \leq R < M \leq N$. For $n \geq 1$, $1 \leq r \leq R$, let $c(n,r) \in \mathbb{C}$, and put $d_0(n) = \sum_r |c(n,r)|$. Define*

$$m_N(n,r) = \big[\min(N-1, n-1) - \max(0, n+r-M) + 1\big]_+,$$

$$B = \frac{1}{MN} \sum_{n,r} m_N(n,r) c(n,r), \qquad C = \frac{1}{T} \sum_{n=1}^{N} \sum_r c(n,r).$$

Then

$$|B - C| \leq \frac{1}{MN} \sum_{n=M}^{N} \sum_r r|c(n,r)| + \frac{1}{N} \left(\sum_{n=1}^{M-1} d_0(n) + \sum_{n=N+1}^{N+M-1} d_0(n) \right) + \left| \frac{1}{N} - \frac{1}{T} \right| \sum_{n=1}^{N} d_0(n). \tag{12}$$

In particular if $d_0 \leq D_0$ everywhere and $\sum_r r|c(n,r)| \leq D_1$ for $M \leq n \leq N$, then

$$|B - C| \leq D_1/M + 2(M-1)D_0/N + D_0/T. \tag{13}$$

Proof. The block $\{t+1, \ldots, t+M\}$ contains both $n, n+r$ exactly when $n + r - M \leq t \leq n - 1$. Intersect with $0 \leq t \leq N - 1$ to obtain m_N. For $M \leq n \leq N$, $m_N = M - r$. Outside this interior, $0 \leq m_N \leq M$, and contributing positive sites have $n \leq N + M - 1$. Compare first with the N-normalized prefix and then change N to T. The triangle inequality gives (12). Finally $N|N^{-1} - T^{-1}| = (T - N)/T \leq 1/T$. □

In the source, aggregate labels with the same displacement $r = hqd$. The ambient endpoint cutoffs must be independent of the block origin except through the sites themselves. Testing (4) gives L times the signed edge summand in each orientation, hence the divisor $2LMN$, not LMN, in the averaged quadratic form. The source uses $D_0 = (K/L)5^{400 \log L}(8W)^J$ and $R_L = he^{100L+1}$. Its bound follows by $D_1 \leq R_L D_0$ (with an integer ceiling if necessary). The improvement is genuine whenever most absolute edge mass is at shorter displacements; it asserts no such arithmetic concentration without evidence. The right endpoint of an edge may exceed T and is not erased in C.

Proposition 6.2 (Absolute edge-mass certificate)**.** *For a finite directed edge set with complex coefficients c_{xy}, and $|f_x|, |g_y| \leq 1$, deleting any set $\mathcal{E}$ of edges changes $\sum c_{xy} f_x g_y$ by at most $\sum_{(x,y)\in\mathcal{E}} |c_{xy}|$. If incoming and outgoing absolute row masses at a vertex are bounded by $DW(x)$, edges incident to a deleted vertex set Z cost at most $2D \sum_{x\in Z} W(x)$. Under a probability average, for $q > 1$,*

$$\mathbb{E}[W \mathbf{1}_Z] \leq (\mathbb{E}W^q)^{1/q} \mathbb{P}(Z)^{1-1/q}. \tag{14}$$

Proof. Apply the triangle inequality, charge edges to a deleted endpoint (double charging only enlarges the bound), and apply Holder's inequality. □

Density alone is insufficient: on a set of probability ε, the weight $W = \varepsilon^{-1}$ has weighted mass one. Formula (14) is valid only where its moment is established. In the qualitative source B is fixed and a finite full period gives moments of W. In the quantitative integer model, one must first use exact low-degree Q states and the appropriate finite-law comparison; unrestricted product-law moments cannot simply be relabeled integer moments. A block-dependent new truncation also changes the word expansion and its circuit/forest obligations.

7 The Qualitative Normalization and Its Sensitivity

The qualitative source starts afresh: $\epsilon = 10^{-4}$, $\eta = \epsilon/100$, $\rho = 1/20$, $\kappa = 400/\eta$, $A = e^{2\kappa}$, $c_* = \epsilon/100$. Its core and center bands are

$$\mathcal{C} = \{p : B^{1-\eta} < \log p \le B\}, \quad \mathcal{Z} = \{p : B^{1-\epsilon} < \log p \le B^{1-\eta}\}.$$

These A, η are not those of Section 3. Put $A_C = A$, $A_Z = 1$, $\beta_a = (1 + A_a)^{-1}$, and $\theta = \beta_Z^2 = 1/4$. For admissible squarefree labels $H < d \le \tau H$, $1 < \tau < 2$, with $\omega(d) \le \lceil C_0 \log B \rceil$, the kernel is

$$g_d(x, x \pm hd) = w_d(x) w_d(x \pm hd) A^{\omega_C(d)} \mathbf{1}_{d_C | x} \prod_{p|d,\, p \in \mathcal{Z}} (\mathbf{1}_{p|x} - \theta/p).$$

Here $w_d \in [0, 1]$ depends only on core residues and has the source lower core-degree support. The vertex normalization is $W(x) = \prod_a \beta_a^{-\omega_a(x)}$, with $L_0 = \mathbb{E}W = \prod_a \prod_{p \in \mathcal{P}_a} (1 + A_a/p)$. The factors $\mathbf{1}_{p|x} - \theta/p$ are *not* individually mean zero.

Proposition 7.1 (Exact centering calibration)**.** *Let p be a prime, $a_0 \in \mathbb{Z}/p\mathbb{Z}$, $0 < \beta < 1$, $\theta \ge 0$, and let $\nu(a)$ be nonnegative integers with $\nu(a_0) = m \ge 1$ and $\sum_a \nu(a) = \ell$. Define*

$$J_p = \frac{1}{p} \sum_{a \bmod p} (\mathbf{1}_{a=a_0} - \theta/p) \beta^{\nu(a)}.$$

Then the exact identity and error bound are

$$J_p = \frac{\beta^m - \theta}{p} + \frac{\theta}{p^2} \sum_a (1 - \beta^{\nu(a)}), \tag{15}$$

$$0 \le J_p - \frac{\beta^m - \theta}{p} \le \frac{\theta(1-\beta)\ell}{p^2}. \tag{16}$$

Thus choosing $\theta = \beta^m$ cancels the order-$1/p$ term. With $\theta = \beta^m + \Delta \ge 0$, its absolute contribution is at most $|\Delta|/p + \theta(1-\beta)\ell/p^2$.

Proof. Separate the distinguished residue and write $\sum_a \beta^{\nu(a)} = p - \sum_a (1 - \beta^{\nu(a)})$. For an integer $j \ge 0$, $1 - \beta^j \le j(1-\beta)$; summing proves the bound. □

The source good singleton has exactly two departures on the lit residue, including all return factors, so $m = 2$ and $\beta = 1/2$. The proposition reproduces its cancellation and supplies an explicit tolerance budget for perturbed centering or normalization. If a family permits two lit multiplicities $m_1 \ne m_2$, a single θ has

$$\max(|\beta^{m_1} - \theta|, |\beta^{m_2} - \theta|) \ge \tfrac{1}{2} |\beta^{m_1} - \beta^{m_2}|.$$

Consequently, with bounded ℓ and $p \to \infty$, one cannot uniformly obtain $O(p^{-2})$ for both profiles by that single centering. This explains why the source isolates good edges and charges nongood ones separately. An edge-dependent calibration could repair a *fixed word*, but is not automatically an admissible original graph kernel. Its analytic uncentering, record counting, and conditional cylinder dependence would need new proofs.

A small Boolean refinement. For a Boolean survival function G on $s \geq 1$ singleton overwrite choices, the full mixed difference has 2^{s-1} positive and 2^{s-1} negative terms, each in $\{0, 1\}$; hence $|\Delta G| \leq 2^{s-1}$ instead of 2^s. For $s = 0$ the bound is one. This halves a crude majorant but does not change any trace exponent. It does not permit dropping the common-configuration witness argument, conditioning on free singleton residues, or taking absolute values before integration.

8 Finite Ram Reformulations With Explicit Residuals

The useful part of private TR190-GGRT [2] is its finite spectral calculus, Parseval reconstruction, and degree-polynomial locality. We neither use its conjectural symbol relations nor infer finite-hop support for an arbitrary small-scale spectral multiplier. Exact finite-hop support below requires a polynomial. Private TR1674 [3] supplies finite Gram and energy identities with explicit analytic-status boundaries.

8.1 Spectral panels do not create cancellation

Let $A = E(\sum_d A_d)E$ on one block and let $A = U\Lambda U^*$. For filters κ_j satisfying $\sum_j |\kappa_j(\lambda)|^2 = 1$ on the finite spectrum,

$$\|f\|^2 = \sum_j \|\kappa_j(A)f\|^2, \qquad \langle f, Af\rangle = \sum_j \langle \kappa_j(A)f, A\kappa_j(A)f\rangle. \tag{17}$$

Diagonalization proves both identities. If the jth filter is supported on $|\lambda| \leq t_j$, the absolute form is at most $\sum_j t_j \|\kappa_j(A)f\|^2$. This is a useful certificate only if the panel energies are bounded independently; Parseval alone gives no saving. For a non-tight frame the first equality must be replaced by its frame bounds. For a unitary change of block coordinates U, use $\mathcal{U} = \mathrm{diag}(U, \ldots, U)$ on the edge-copy space:

$$H' = \mathcal{U}^* H \mathcal{U}, \qquad \|(H')^k\|_{\mathrm{HS}} = \|H^k\|_{\mathrm{HS}}, \quad \rho(H') = \rho(H).$$

This is an exact reformulation of the high trace, not a smaller trace estimate.

For a directed complex matrix M in the qualitative source, the correct self-adjoint lift is

$$\mathcal{A} = \begin{pmatrix} 0 & M \\ M^* & 0 \end{pmatrix}, \qquad \mathrm{Tr}(\mathcal{A}^{2m}) = 2\,\mathrm{Tr}((M^*M)^m), \quad \|\mathcal{A}\| = \|M\|.$$

The identities follow by block multiplication and the equality of nonzero singular values. They preserve the factor two. For an unconjugated correlation $\sum c_{xy}F(x)G(y)$ with $M_{xy} = c_{xy}/\sqrt{W(x)W(y)}$, the test vectors are $u(x) = \overline{F(x)}\sqrt{W(x)}$ and $v(y) = G(y)\sqrt{W(y)}$; then $\langle u, Mv\rangle$ is exactly that correlation, with the endpoint supports kept.

Proposition 8.1 (Polynomial restriction residual)**.** *Let A be self-adjoint on a finite subset of $\mathbb{Z}$, with $A_{xy} = 0$ for $|x - y| > R$ and $\|A\| \leq D$. Let P restrict to an interval I, $A_I = PAP|_{\ell^2(I)}$, and*

$p(z) = \sum_{j=0}^{s} c_j z^j$. *For* $f \in \ell^2(I)$, *let* f_∂ *be its restriction to sites whose distance to the ambient complement of* I *is at most* sR. *With zero extensions understood,*

$$\|p(A)f - p(A_I)f\| \le 2\left(\sum_{j=1}^{s} |c_j| D^j\right) \|f_\partial\|. \tag{18}$$

Proof. For an input coordinate farther than sR from the complement, no matrix path of length at most s leaves I, so the two polynomial actions agree there as outputs of that input. The constant terms agree on all inputs. Apply the triangle inequality and $\|A_I\| \le \|A\| \le D$ to f_∂. □

Compression and functional calculus do not generally commute. For $A = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and projection onto the first coordinate, $PA^2P = P$ but $(PAP)^2 = 0$. The proposition therefore supplies a boundary residual that a bare transform claim would miss. It applies to finite-range adjacency; substituting $\rho(H)$ for a norm bound on the nonnormal H is invalid.

8.2 A source-specific energy interface and its quantifier gap

For $X \in \mathbb{N}$, let $C_{nj} = \mathbf{1}_{j \le n}$, $W_X = \operatorname{diag}(1/[n(n+1)])$, and $K_X = C^* W_X C$. For real vectors z,

$$\sum_{n \le X} \frac{|\sum_{j \le n} z_j|^2}{n(n+1)} = z^* K_X z, \qquad (K_X)_{ij} = \frac{1}{\max(i,j)} - \frac{1}{X+1}. \tag{19}$$

The kernel formula follows by telescoping, so the terminal correction is essential. For $z = \lambda$, set $A_h(y) = \sum_{n \le y} \lambda(n)\lambda(n+h)$ and

$$\delta_h(X) = \max_{1 \le y \le X-h} \frac{|A_h(y)|}{y} \quad (1 \le h < X).$$

The maximum is over integer y. Write $H_X = \sum_{n \le X} 1/n$.

Proposition 8.2 (All-shift profile certificate)**.** *For every* $X \ge 2$,

$$V(X) := \lambda^* K_X \lambda \le H_X - \frac{X}{X+1} + 2\sum_{h=1}^{X-1} \delta_h(X) k_h(X), \tag{20}$$
$$k_h(X) = H_X - H_h - \frac{X-h}{X+1} \ge 0.$$

Thus any certified upper profiles $d_h(X) \ge \delta_h(X)$ *with* $\sum_{h<X} d_h(X)k_h(X) \le B(X)$ *imply* $V(X) \le H_X - X/(X+1) + 2B(X)$.

Proof. Expand (19) into the diagonal and its two equal off-diagonal parts. For fixed h, the latter sum is $\sum_{n=1}^{m} \lambda(n)\lambda(n+h)v_n$, where $m = X - h$ and $v_n = 1/(n+h) - 1/(X+1) > 0$ is decreasing. Summation by parts bounds its absolute value by

$$\delta_h(X)\left[m v_m + \sum_{n=1}^{m-1} n(v_n - v_{n+1})\right] = \delta_h(X)\sum_{n=1}^{m} v_n = \delta_h(X)k_h(X).$$

Summing proves the claim. □

This deliberately exposes a limitation: $y = 1$ already gives $\delta_h(X) = 1$ for Liouville, so this basic supremum certificate is not itself a cancellation bound. A useful version must split off small y and use a genuinely scale-dependent prefix profile in the same summation-by-parts formula. More importantly, (1) fixes h before taking $X \to \infty$ and gives no uniform estimate for all $h < X$. It therefore does not by itself bound the off-diagonal energy sum. With trivial profiles,

$$\sum_{h<X} k_h(X) = X - H_X - \frac{X(X-1)}{2(X+1)} = O(X),$$

not $O(\log X)$. No RH conclusion, quarter-scale contraction conclusion, or uniform all-shift assertion is inferred. TR1674's carry term, cross term, noise term, and boundary term must remain separate; Gram positivity cannot remove the last three.

9 Conditional Assembly and Every-Cutoff Obligations

In the quantitative route the source has, at *each* sufficiently large real X, the normalized centering and deletion errors

$$\eta + X^{-1} + \eta^{-1}2^J L^{-1/20}, \qquad 2^J(K^{-1} + L^{-100} + e^{-2WJ}) + e^{-L^{0.9}}.$$

If these inputs and (5) hold, substituting either optimized threshold testing or (11) changes only the finite spectral transfer stage. The $O(L/\eta)$ bins cancel the $1/L$ in each bin estimate; division by $S_0 \geq SV_*/2$ retains the essential W^J denominator. An absolute spectral constant is fixed before selecting W large enough, then the cutoff threshold may depend on h, l, W. The source exponent $c = \delta/(6WA)$ is not improved here without redoing this complete choice of constants. Exceptional *block origins* at one cutoff are not exceptional *cutoffs*; their mass is paid at every X.

The quantitative affine identity also preserves all endpoints. With $l = a_1a_2$, $c_0 = \min(a_2b_1, a_1b_2)$ and $h = |a_1b_2 - a_2b_1|$, let $P(Y) = \sum_{1\leq m\leq Y,\, m\equiv c_0 \pmod{l}} \lambda(m)\lambda(m+h)$. Then exactly

$$\sum_{1\leq n\leq X} \lambda(a_1n + b_1)\lambda(a_2n + b_2) = \lambda(l)\{P(lX + c_0) - P(c_0)\}.$$

No unit-class assumption is permitted. For general multiplicative functions one instead needs the source finite local expansions, since $f(p^k) = f(p)^k$ need not hold. The read expansion $F_{a,f}(m) = \mathbf{1}_{a|m} f(m/a)$ is a signed sum of $2^{\omega(a)}$ normalized 1-bounded multiplicative functions differing from f only at primes dividing a. Each inherits (2); one cannot place the nonpretentiousness hypothesis only on a conveniently chosen product.

In the qualitative route the imported high trace controls $\sum_{\boldsymbol{d}} |\mathbb{E}K_{\boldsymbol{d}} \prod_i \mathbf{1}_Y|$ by $B^{-(1+c_*)\ell}$ and the deleted density by $P_0^{-1+o(1)}$, with $\ell = 2\lfloor B \rfloor$ and $P_0 = e^{B^{1-\epsilon}}$. The directed block matrix uses singular moments $\mathrm{Tr}((M^*M)^{\ell/2})$, not $\mathrm{Tr}(M^\ell)$. A power-of-two block length $D_0 \asymp B^{10}H$ is coprime to the odd-prime period Q_B. This gives independence of block position and prime residue *over the full period*, not independence of the exceptional-block event and W. The latter is controlled by Cauchy–Schwarz.

For each fixed B the long ordinary average tends to its periodic comparison; only afterwards does B tend to infinity. The analytic contradiction is $c_\gamma \leq C(B^{-c_*/2} + B^{-\epsilon} + BJ/P_0)$. This two-stage limit excludes every alleged biased subsequence if all named inputs hold. It supplies no prescribed convergence rate under mere (2). The unread qualitative cylinder, thinning, final enumeration, and rough-number proofs remain explicit imported obligations.

10 Public Context, Tests, and Assessment

10.1 Current primary literature, checked October 8, 2026

Public queries used the public source title and public mathematical title, author, and topic identifiers only. No private text, private title, or draft was transmitted. Primary records, not secondary search summaries, support the following distinctions. Matomaki–Radziwill–Tao [4] provide averaged short exponential sums; the checked record is the corrected March 2022 version. Tao [5] proves logarithmically averaged two-point cancellation. Helfgott–Radziwill [6] obtain graph expansion and quantitative logarithmic cancellation. Pilatte's August 25, 2026 version [7] gives a power-logarithmic saving still with logarithmic weight. Tao–Teravainen's April 25, 2026 version [8] gives quantitative correlations; the source uses its exceptional-scale comparison, not an every-cutoff replacement.

A current search additionally found Guo's revised September 1, 2026 title *Logarithmic Chowla Correlations Across All Shift Scales* [9]. Its abstract claims full shift-range logarithmic bounds outside a common exceptional set and a separate GRH-conditional statement. The earlier search snippet carried an older title; the direct current version was checked. This is recorded as an abstract-level literature update, not a proof read or an imported theorem. These public comparisons do not establish historical priority of the local resolvent inequality. The OpenAI primary release [1] is the source being studied; neither its release status nor its separate formalization description is a proof audit performed here.

10.2 Verification and limitations

The accompanying executable tests passed 24,617 finite checks. They use rational arithmetic for finite edge window multiplicities, calibration identities, mixed differences, and arithmetic energy identities. Seeded finite complex Hermitian matrix examples check the noncommuting transfer, weighted-tail certificate, Parseval/lift identities, and polynomial boundary residual with documented floating tolerance. They include noncommuting edge pairs, projections not commuting with the edge matrices, and zero compressed-square cases. Manufactured finite examples are not arithmetic-scale experiments on the source graph. No test of finitely many cutoffs verifies (1), the forest count, or asymptotic nonpretentiousness. The parent's 2,640 scalar parameter checks of the optimized bound are separately attributed, not recounted as author tests or global proof.

The native editor/compiler is attempted first. The installed export helper provides the requested on-disk PDF if the native platform path fails. The final build ledger records compilation, every-page visual inspection, and hashes; these establish artifact integrity and presentation, not theorem validity. At most two cosmetic build rounds are allowed.

10.3 Advantages, costs, and research directions

The optimized transfer preserves all noncommutation and projection issues and can substantially reduce a finite transfer constant. Weighted-tail integration keeps energy concentration visible and can replace a fixed exceptional-block split. Displacement-mass bounds improve a boundary estimate when actual edge lengths justify it. The calibration formula identifies exactly where a normalization error restores an order-$1/p$ term. Finite Ram panels and Gram forms provide interchangeable certificates with explicit boundary residuals.

The costs are equally important: source-sized matrices are enormous; changing basis preserves the difficult moment; a smaller local bound need not improve the final exponent; and word-dependent calibration is not automatically compatible with the analytic kernel. Local polynomial

approximations to Liouville require residuals, not assumed smoothness. The C2 contribution is therefore limited but concrete, independent of L3 applicability.

Useful next research would make C_2 and the cutoff thresholds explicit; develop anisotropic square certificates without changing the prime residue integration order; optimize the integrated tail with the exact Ψ instead of its piecewise linear envelope; and establish weighted displacement moments for retained arithmetic edges. A scale-dependent all-shift energy profile would be a separate substantial arithmetic input. Any modification of the deletion predicate must revalidate finite circuit complexity, common-hybrid witnesses, and a forest-code universe uniform over all padding coefficients. None of these tasks is claimed completed by this report.

References

[1] OpenAI, *Ordinary two-point correlations of multiplicative functions*, September 24, 2026. Pinned build at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`; parent captured-target identity at `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`. Primary release and citation record.

[2] M. Subbarao, private TR190-GGRT, *The Graph-Global Ram Transform*, actual catalogue-route original, selected passages as in Section 2. Local confidential reference; no public upload.

[3] M. Subbarao, private TR1674, revised-v2, finite arithmetic energy and Gram formulation, selected passages as in Section 2. Local confidential reference; analytic claims beyond the finite identities are not imported as local theorems.

[4] K. Matomaki, M. Radziwill, and T. Tao, *An averaged form of Chowla's conjecture*, Algebra Number Theory 9 (2015), 2167–2196; corrected v3, March 1, 2022. `https://arxiv.org/abs/1503.05121v3`.

[5] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi 4 (2016), e8. `https://arxiv.org/abs/1509.05422v4`.

[6] H. A. Helfgott and M. Radziwill, *Expansion, divisibility and parity*, 2021. `https://arxiv.org/abs/2103.06853`.

[7] C. Pilatte, *Improved bounds for the two-point logarithmic Chowla conjecture*, v3, August 25, 2026. `https://arxiv.org/abs/2310.19357v3`.

[8] T. Tao and J. Teravainen, *Quantitative correlations and some problems on prime factors of consecutive integers*, v2, April 25, 2026. `https://arxiv.org/abs/2512.01739v2`.

[9] J. Guo, *Logarithmic Chowla Correlations Across All Shift Scales*, v4, September 1, 2026. Abstract-level update only. `https://arxiv.org/abs/2608.23500v4`.